variation of polarizability can be explained on the basis of the relaxation times of the various contributing polarization processes. When the frequency of the applied field is quite large as compared to the inverse of the relaxation time for a particular polarization process, the contribution of that process to polarizability is negligible. As relaxation time is maximum for the dipolar process and minimum for the electronic process, the dipolar contribution disappears first followed by ionic and electronic contributions.

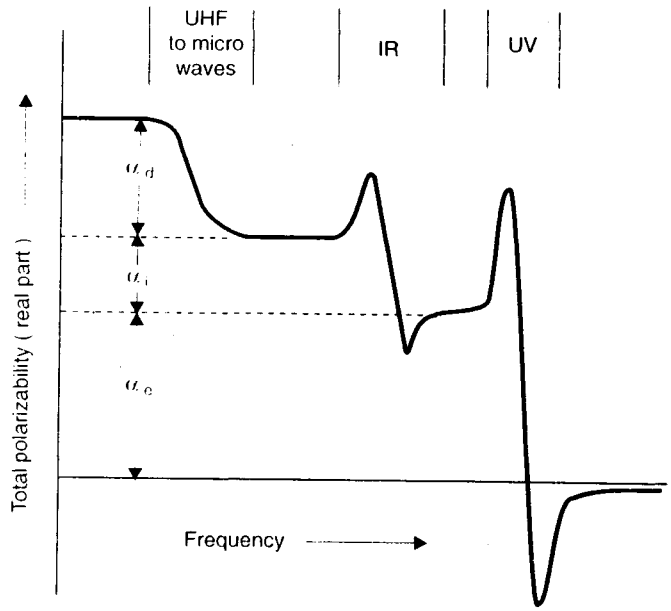


Fig. 9.4. *Frequency dependence of various contributions to polarizability.*

9.6 FERROELECTRICITY

Ferroelectricity is the phenomenon which refers to the state of *spontaneous polarization,* i.e., polarization of the meterial in the absence of an electric field. It is thus analogous to ferromagnetism which represents the state of spontaneous magnetization of the material. The crystals exhibiting ferroelectricity are called the *ferroelectric crystals.* In such crystals, the centres of positive and negative charges do not coincide with each other even in the absence of the field, thus producing a non-zero value of the dipole moment. The variation of polarization with electric field is not linear for such crystals but forms a closed loop called the *hysteresis loop.* The ferroelectricity disappears above a certain critical temperature called the *transition temperature* or the *Curie point,* T_C, when the material gets transformed from ferroelectric to paraelectric state as indicated by a rapid decrease in the dielectric constant

with increase in temperature.

Ferroelectricity is observed in a number of substances; the most familiar ones are barium titanate ($BaTiO_3$) and the Rochelle salt ($NaKC_4H_4O_6 \cdot 4H_2O$). The ferroelectric crystals may be classified into two main groups—the *order-disorder group* and the *displacive group*. In the order-disorder class of ferroelectrics, the ferroelectric transition is associated with individual ordering of ions. These are the crystals which contain hydrogen bonds and in which the motion of protons is related to the ferroelectric properties. The examples are potassium dihydrogen phosphate (KH_2PO_4), rubidium hydrogen phosphate (RbH_2PO_4), etc. The displacive group of ferroelectrics is the one in which the ferroelectric transition is associated with the displacement of a whole sublattice of ions of one type relative to a sublattice of another type. The crystals of this class exhibit structures which are closely related

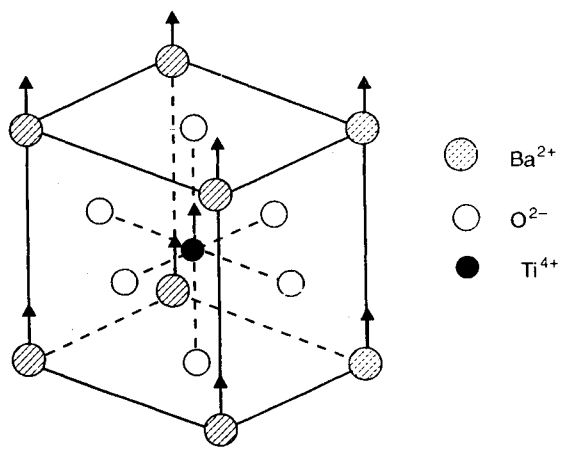


Fig. 9.5. *Structure of $BaTiO_3$ for $T > T_C$.*

to the perovskite and ilmenite structures. The examples are $BaTiO_3$, $KNbO_3$, $LiTaO_3$, etc. Consider the case of $BaTiO_3$ crystal. Below the Curie point (380 K), it exhibits the perovskite structure as shown in Fig. 9.5. The unit cell is cubic with Ba^{2+} ions occupying the corners, O^{2-} ions occupying the face centres and Ti^{4+} ion occupying the body centre of the cube. Thus each Ti^{4+} ion is surrounded by six O^{2-} ions in an octahedral configuration. For $T > T_C$, the centres of gravity of positive and negative charges exactly coincide with each other to produce no net dipole moment. However, for $T < T_C$, the Ti^{4+} and Ba^{2+} ions slightly move upwards while the O^{2-} ions slightly move downwards and the structure becomes tetragonal with centres of positive and negative charges not coinciding with each other. This produces the spontaneous polarization in the crystal.

In fact, the ferroelectricity in $BaTiO_3$ arises from induced electronic and ionic dipole moments. Rewriting Eq. (9.13) for the dielectric constant, we get

$$\varepsilon_r = \frac{1 + \frac{2}{3\varepsilon_o} \sum_j N_j \alpha_j}{1 - \frac{1}{3\varepsilon_o} \sum_j N_j \alpha_j} \tag{9.33}$$

where α_j is the electronic plus ionic polarizability of the jth ion and N_j is the number of jth ions per unit volume. The expression (9.33) has been obtained by assuming that the local field at all the atoms is equal to $\mathbf{E} + \frac{\mathbf{P}}{3\varepsilon_o}$. From Eq. (9.33), we find that, for

$$\sum_j N_j \alpha_j = 3\varepsilon_o, \tag{9.34}$$

the dielectric constant becomes infinite which permits a finite polarization in the absence of the applied field. This is the condition for the so-called *polarization catastrophe*. In polarization catastrophe, the local electric field arising from polarization increases faster than the elastic restoring force on an ion in the crystal, thereby producing an asymmetrical shift in ionic position. The shift is, however, limited to finite displacements by the higher order restoring forces.

In Eq. (9.34), the value of ε_r is sensitive to small departures of $\Sigma N_j \alpha_j$ from the critical value of $3\varepsilon_o$. Putting

$$\frac{1}{3\varepsilon_o} \sum_J N_j \alpha_j = 1 - 3s, \tag{9.35}$$

where $s << 1$, in expression (9.33), the dielectric constant becomes

$$\varepsilon_r = \frac{1}{s} - 2 \cong \frac{1}{s} \tag{9.36}$$

We assume that s varies linearly near the critical temperature as

$$s \cong \frac{T - T_C}{\xi} \tag{9.37}$$

where ξ is a constant. This type of variation of s or $\sum_j N_j \alpha_j$ might result from normal thermal expansion of the lattice. From Eq. (9.36), we obtain

$$\varepsilon_r \cong \frac{\xi}{T - T_C} \tag{9.38}$$

Such a variation of dielectric constant with temperature is close to that observed experimentally in the paraelectric state.

9.7 PIEZOELECTRICITY

In cetain crystals, the application of an external stress induces a net dipole moment which produces the electric polarization with the polarization charges appearing on the surfaces of the crystals. Such crystals are called the *piezoelectric crystals* and the phenomenon is known as the *piezoelectricity.* Some examples of such crystals are quartz, the Rochelle salt and tourmaline. The inverse effect is also observed, i.e., the application of an electric field produces strains in the crystal.

In schematic one-dimensional notation, the piezoelectric equations are

$$\left.\begin{aligned} P &= Zd + \varepsilon_0 \, E \, \chi \\ e &= Zs + Ed \end{aligned}\right\} \tag{9.39}$$

where P represents the polarization, Z the stress, d the piezoelectric strain constant, E the electric field, χ the dielectric susceptibility, e the elastic strain and s the elastic compliance constant. The first one of the equations (9.39) exhibits the development of polarization by an applied stress and the second one shows the development of elastic strain by an applied electric field. Generally, very large electric fields are needed to produce very small strains. In quartz, for example, an electric field of about 10^4 Vm^{-1} produces a strain of about 1 in 10^8 only.

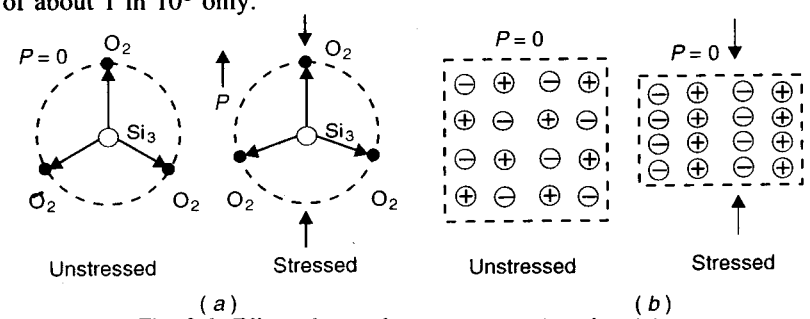


Fig. 9.6. *Effect of crystal symmetry on piezoelectricity.*
(a) Quartz crystal with no centre of inversion shows piezoelectricity.
(b) A crystal with centre of inversion shows no piezoelectricity.

The occurrence of piezoelectricity is the result of displacement of ions in certain crystals under the effect of the applied stresses. In such crystals, the ions are so displaced that their charge distribution loses the original symmetry as shown in Fig. 9.6a. In certain other crystals (Fig. 9.6b), where the symmetry of the charge distribution is not disturbed even after distortion, no piezoelectricity is observed. The latter type of crystals are those which possess the *centre of inversion.* Thus the absence of the centre of inversion

is the pre-requisite for the occurrence of piezoelectricity. It may also be noted that all ferroelectric crystals are piezoelectric but all piezoelectric crystals are not necessarily ferroelectric. An example of the former type of crystals is barium titanate and that of the latter type of crystals is quartz.

Piezoelectric materials are used to convert electrical energy into mechanical energy and vice versa, i.e., they act as electro-mechanical transducers. They are used in devices such as gramophone pick-ups, microphones, strain gauges, ultrasonic generators, etc.

SOLVED EXAMPLES

Example 9.1. Determine the percentage of ionic polarizability in the sodium chloride crystal which has the optical index of refraction and the static dielectric constant as 1.5 and 5.6 respectively.

Solution. From the Clausius-Mossotti relation, we have

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N(\alpha_e + \alpha_i)}{3\varepsilon_0} \tag{9.40}$$

Here α_e and α_i are the electronic and ionic contributions to polarizability respectively. At optical frequencies, Eq. (9.40) becomes

$$\frac{n^2 - 1}{n^2 + 2} = \frac{N\alpha_e}{3\varepsilon_0}, \tag{9.41}$$

where n is the index of refraction. From Eqs. (9.40) and (9.41), we obtain

$$\left(\frac{n^2 - 1}{n^2 + 2}\right)\left(\frac{\varepsilon_r + 2}{\varepsilon_r - 1}\right) = \frac{\alpha_e}{\alpha_e + \alpha_i} \tag{9.42}$$

The right hand side of Eq. (9.42) is the fraction of electronic polarizability. Thus the percentage of ionic polarizability is

$$\left(\frac{\alpha_i}{\alpha_e + \alpha_i}\right) \times 100 = \left(1 - \frac{\alpha_e}{\alpha_e + \alpha_i}\right) \times 100$$

$$= \left[1 - \left(\frac{n^2 - 1}{n^2 + 2}\right)\left(\frac{\varepsilon_r + 2}{\varepsilon_r - 1}\right)\right] \times 100$$

$$= \left[1 - \left\{\frac{(1.5)^2 - 1}{(1.5)^2 + 2}\right\}\left\{\frac{5.6 + 2}{5.6 - 1}\right\}\right] \times 100$$

$$= 51.4\%$$

SUMMARY

1. A dielectric material gets polarized in the presence of an external electric field **E**. The polarization **P** is defined as the dipole moment per unit volume of the material and is related to the electric susceptibility, χ_e, as

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$$

2. The electric displacement vector **D** for an isotropic or cubic medium is defined as

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon_0 \mathbf{E} + \mathbf{P}$$

where ε_r is the relative permittivity or dielectric constant of the dielectric and is related to the electric susceptibility as

$$\varepsilon_r = 1 + \chi_e$$

3. The electric field acting at the site of an atom or molecule is called the local field. This is responsible for polarization of the individual atoms or molecules in the solid. For an atomic site with cubic symmetry, it is given by the Lorentz equation :

$$\mathbf{E}_{loc} = \mathbf{E} + \frac{\mathbf{P}}{3\varepsilon_0}$$

4. The atomic polarizability, α, is defined as the dipole moment, p, per unit local electric field at an atom and is given by the relation :

$$p = \alpha E_{loc}$$

5. The polarizability is an atomic property whereas the dielectric constant is a macroscopic property of the material. The two parameters are related to each other by the Clausius-Mossotti relation given as

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$$

6. The net polarizability of a dielectric material results from three main contributions — the electronic polarizability, ionic polarizability, and dipolar or orientational polarizability. At low frequencies of the applied electric field, all the three contributions are present. As the frequency increases, the dipolar contribution disappears first followed by ionic and electronic contributions at higher frequencies. At optical frequencies, the polarization is mainly due to electronic contribution.

7. Ferroelectric materials are those which exhibit spontaneous polarization, i.e., polarization in the absence of an applied field. These are analogous to ferromagnetic materials. They possess a Curie point and exhibit hysteresis loop. The examples are barium titanate, the Rochelle salt, etc.

8. Piezoelectric crystals are those which are polarized under the application of an external stress. The examples are quartz, the Rochelle salt, tourmaline, etc. The inverse effect is also observed, i.e., the application of an electric field produces strains in such crystals. The absence of the centre of inversion is the pre-requisite for the occurrence of piezoelectricity.

9. All ferroelectric crystals are piezoelectric, but all piezoelectric crystals are not necessarily ferroelectric. Examples of ferroelectric and piezoelectric crystals are barium titanate and quartz respectively.

VERY SHORT QUESTIONS

1. Define the following terms:

   Polarisation, electric susceptibility, local field, dielectric constant, electric displacement vector, polarisability, spontaneous polarisation.
2. Enlist the various contributions to total polarisability.
3. Explain the following in brief:

   Ionic polarisation, electronic polarisation, orientational polarisation.
4. Which type of polarisation is the most effective in the visible region?
5. What is ferroelectricity? Give an example of ferroelectric crystal.
6. What is piezoelectricity?
7. Give an example of a crystal that is piezoelectric but not ferroelectric.
8. How is dielectric constant related to electric susceptibility ?

SHORT QUESTIONS

1. Derive the Clausius-Mossotti relation expressing the relationship between dielectric constant and atomic polarisability.
2. What is dipolar polarisability? Obtain an expression for dipolar polarisability of a dielectric at moderate temperatures.
3. What is electronic polarisability? Derive an expression for electronic polarisability using the classical theory.
4. Why is ionic polarisability found to be rather insensitive to temperature ? Give a possible explanation.
5. How does the total polarisability depend on frequency?
6. Why do piezoelectric crystals having centre of inversion show no piezoelectricity?
7. Describe the characteristic properties of ferroelectric materials. What is meant by polarisation catastrophe?

8. Explain the meaning and origin of piezoelectricity. Justify the statement " All ferroelectric crystals are piezoelectric, but all piezoelectric crystals are not necessarily ferroelectric.

PROBLEMS

1. The optical index of refraction and the dielectric constant for water are 1.33 and 8.1 respectively. Determine the percentage of ionic polarizability. (71%)

2. Calculate the polarization of a $BaTiO_3$ crystal using the structure as shown in Fig. 9.5. The structure transforms from cubic to tetragonal below the Curie point with lattice parameters c and a equal to 4.03Å and 3.98 Å respectively. The displacement of titanium ion is opposite to that of oxygen ions. The magnitude of displacement is 0.06 Å for titanium, 0.06 Å for oxygen ions on the side faces and 0.08 Å for oxygen ions on the top and bottom faces of the cube. The displacement of barium ions may be neglected. ($0.16\ cm^{-2}$)